

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Forward Deployed Engineer / Applied AI specialist – Production LLM agents (Fortinet RAG, 70% resolution-time reduction) – Agentic frameworks with evaluation suites (CAD 202-test harness, ARTEMIS multiple VLM backends, ATHENA 5-agent orchestration) – Technical advisory & solution architecture for enterprise AI adoption

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning Specialization), **GPA: 4.0/4.0**

Aug 2024 – Dec 2026

– **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system for enterprise network telemetry: LLM plans multi-step tool calls, invokes APIs, and verifies hypotheses via structured function calling to produce autonomous root-cause analysis
- customized pilot from hackathon prototype to production deployment, reducing mean resolution time 70% through cross-functional collaboration with platform and support teams.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection; reduced manual troubleshooting 75% by replacing heuristic rules with learned thresholds.
- Scaled OpenSearch ingestion pipelines from 50 to 2,000 events/sec (40x) using Golang; deployed edge ML inference via ONNX Runtime (40% latency reduction); built SLA forecasting for 60+ classifiers across 4 categories to resolve technical roadblocks in production monitoring.
- Designed evaluation suites for LLM-driven diagnostics: automated test harnesses validating tool-call correctness, hypothesis verification accuracy, and end-to-end resolution quality across diverse network topologies.

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop agentic pipeline for parametric CAD code generation with inference-time search and verifiable geometric reward: frozen LLM generates CadQuery programs, execution feedback and geometric verification (Chamfer/Hausdorff on STL/B-Rep) drive a LangGraph-orchestrated repair loop with structured evidence prompts – 16-module system with 202-test evaluation harness and VLM-based visual judgment for bespoke code-generation solutions.
- Designing geometric verification components scoring compile validity, parametric 3D correctness, and repair trajectory quality; AST/code-graph structural analysis for failure diagnosis and attributed source-edit spans enabling iterative repair under cost/latency constraints.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ARTEMIS** (Aug 2025–Present): Developing cost-aware VLM routing for agentic tool-use selection; trained neural multi-task router with SLA-aware load balancing serving multiple VLM backends (Gemma 3, Qwen-VL, Llama-4) via vLLM with dynamic routing modes (accuracy/cost/latency) – production-grade inference optimization for customized multimodal deployments.
- **SHAstra** (Jan 2026–Present): Designing a trace-to-graph pipeline converting agent execution traces (GAIA, 94 sessions, 4,037 events) into reusable task-composition graphs with dependency/control/conditional edges; LangGraph-based planner with Pydantic state schemas and component registry for compositional workflow reuse – enabling long-horizon task decomposition with evaluation-driven iteration.
- **ATHENA** (Jan–May 2025): Built a multi-agent screenplay generation system with supervisor-worker orchestration via LangGraph Command routing; coordinated 5 specialized agents with reflection-based quality critique and dynamic plan modification; Pydantic-validated structured outputs; evaluated with cross-modal metrics (BLEU-4 +0.18 absolute, CLIPScore, METEOR, BERTScore, SSIM, PSNR) on 100 reference videos.

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection

TECHNICAL SKILLS

Agentic AI: LangChain, LangGraph, Agentic RAG, Tool/Function Calling, Multi-Agent Orchestration, Supervisor-Worker Patterns, Planning & Reasoning, LLM-as-a-Judge, Structured Generation (Pydantic), Workflow Composition, Long-Horizon Task Decomposition, Evaluation Suites

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, CNNs, Vision Transformers, NLP, Computer Vision, Scikit-Learn, NumPy, Pandas

LLMs / VLMs: vLLM, VLM Routing & Evaluation, CLIP, SBERT, FAISS, Inference Optimization, Model Evaluation & Benchmarking, Cross-Modal Retrieval, Cost-Aware Serving

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, Linux, Git, ONNX Runtime, FastAPI, Azure, W&B, TensorBoard